

Clinical Risk Assessment Report: Patient 55

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 14.1% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.64x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Patient Age (Age), which reduced the patient's absolute risk by 2.4%. Biologically, this feature patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

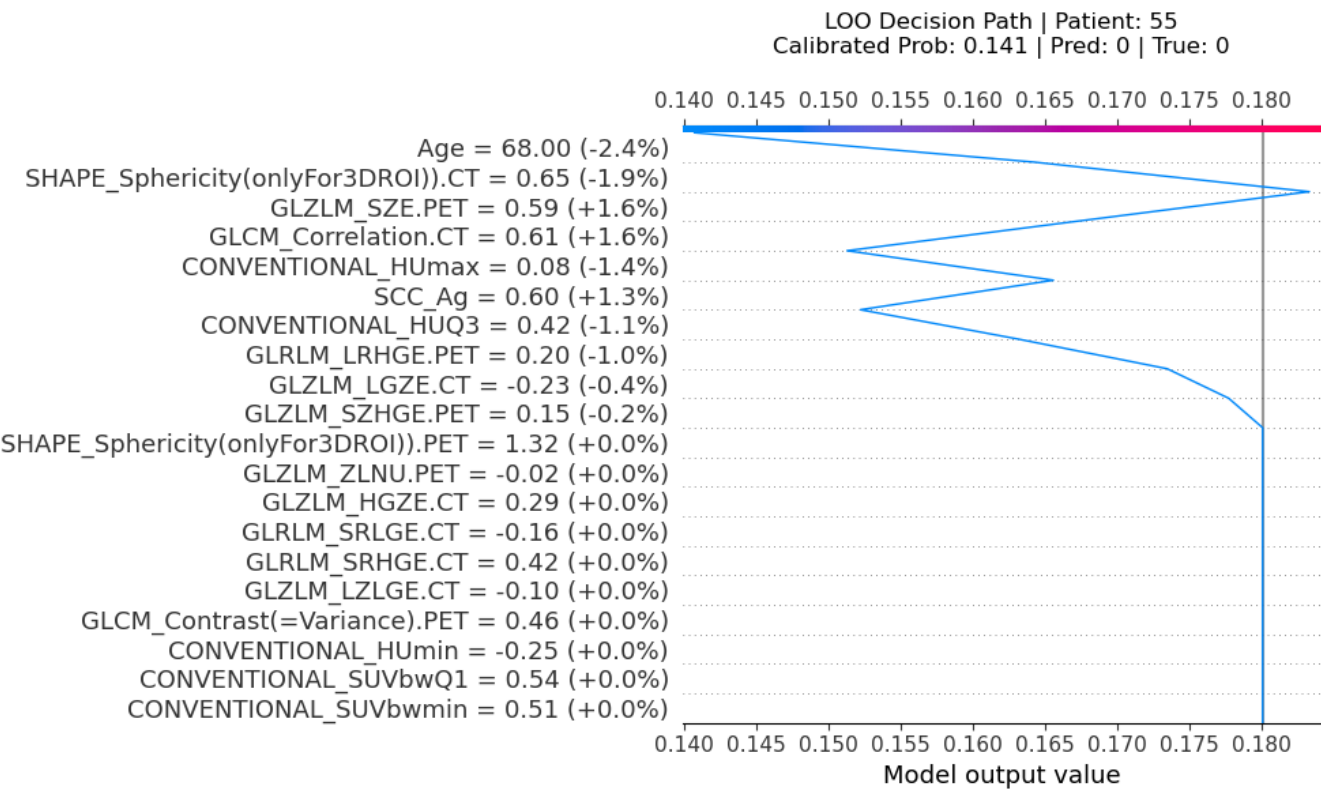
2. Key Pathological & Imaging Drivers (Elevating Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET)	Value: 0.59 [Avg (74th %ile)] (+1.6%)
<i>Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.</i>	
Tumor Pixel Correlation (CT) (GLCM_Correlation.CT)	Value: 0.61 [Avg (72th %ile)] (+1.6%)
<i>Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.</i>	
Squamous Cell Carcinoma Antigen (SCC_Ag)	Value: 0.6 [Bottom 18%] (+1.3%)
<i>Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.</i>	

3. Protective / Mitigating Factors (Reducing Risk)

Patient Age (Age)	Value: 68.0 [Avg (68th %ile)] (-2.4%)
<i>Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.</i>	
Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT)	Value: 0.65 [Avg (74th %ile)] (-1.9%)
<i>Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.</i>	
Maximum Tumor Density (CONVENTIONAL_HUmax)	Value: 0.08 [Avg (48th %ile)] (-1.4%)
<i>Reflects the most dense solid components or calcifications within the primary tumor lesion.</i>	
75th Percentile Tumor Density (CONVENTIONAL_HUQ3)	Value: 0.42 [Avg (55th %ile)] (-1.1%)
<i>Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.</i>	
Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET)	Value: 0.2 [Top 22%] (-1.0%)
<i>Points to continuous, elongated bands of high-density or highly active tumor tissue.</i>	

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.

